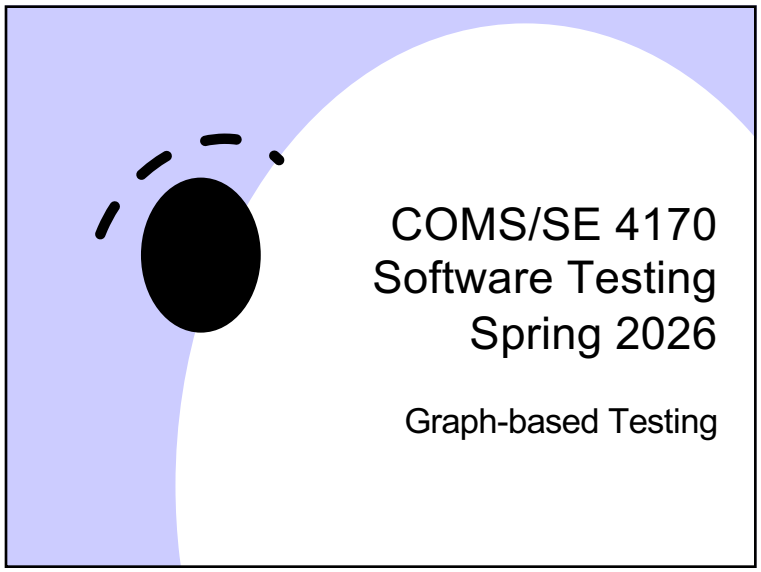
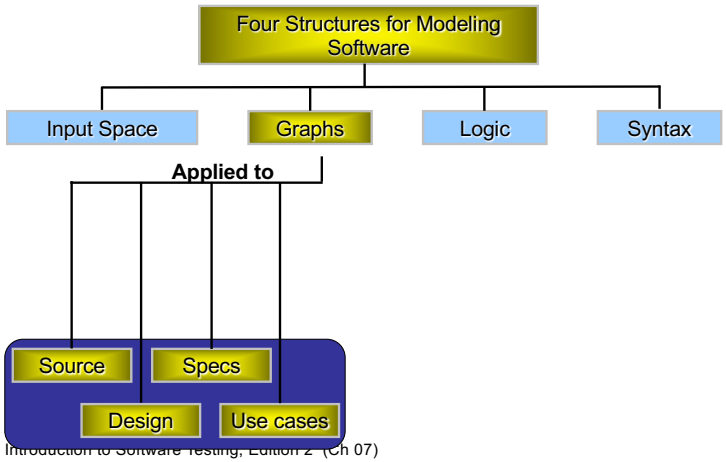


COMS/SE 4170 Software Testing Spring 2026

Graph-based Testing

1

Ch. 7 : Graph Coverage



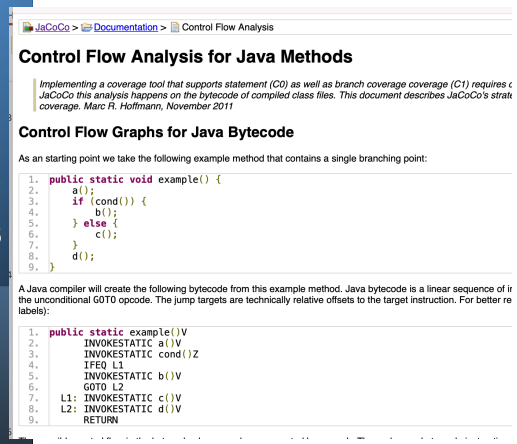
2

Covering Graphs

- Graphs are the most commonly used structure for testing
- Graphs can come from many sources
  - Control flow graphs
  - Design structure
  - FSMs and statecharts
  - Use cases
  - Interfaces (GUIs)
- Tests usually are intended to "cover" the graph in some way

3

Covering Graphs



JaCoCo > Documentation > Control Flow Analysis

Control Flow Analysis for Java Methods

Implementing a coverage tool that supports statement (C0) as well as branch coverage coverage (C1) requires c
JaCoCo this analysis happens on the bytecode of compiled class files. This document describes JaCoCo's strate
coverage. Marc R. Hoffmann, November 2011

Control Flow Graphs for Java Bytecode

As an starting point we take the following example method that contains a single branching point:

```
1.  public static void example() {
2.      a();
3.      if (cond()) {
4.          b();
5.      } else {
6.          c();
7.      }
8.      d();
9.  }
```

A Java compiler will create the following bytecode from this example method. Java bytecode is a linear sequence of ir
the unconditional GOTO opcode. The jump targets are technically relative offsets to the target instruction. For better re
labels):

```
1.  public static example()V
2.        INVOKESTATIC a()V
3.        INVOKESTATIC cond()Z
4.        IFEQ L1
5.        INVOKESTATIC b()V
6.        GOTO L2
7.    L1: INVOKESTATIC c()V
8.    L2: INVOKESTATIC d()V
9.        RETURN
```


4

Definition of a Graph

- A set *N* of nodes, *N* is not empty
- A set N_0 of initial nodes, N_0 is not empty
- A set N_f of final nodes, N_f is not empty
- A set *E* of edges, each edge from one node to another
  - (n_i , n_j), *i* is predecessor, *j* is successor

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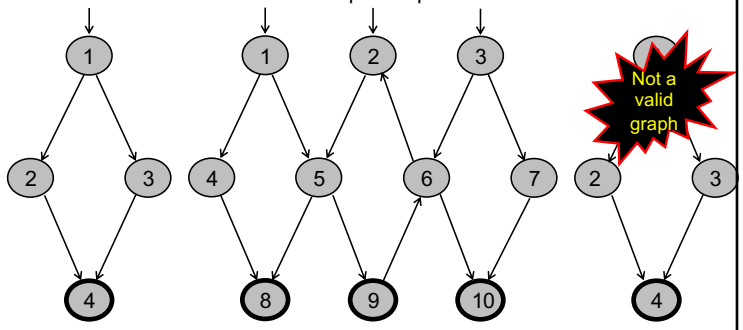
Example Graphs



N_0 = { 1}
N_f = { 4 }
E = { (1,2), (1,3), (2,4), (3,4) }

N_0 = { 1, 2, 3 }
N_f = { 8, 9, 10 }
E = { (1,4), (1,5), (2,5), (3,6), (3, 7), (4, 8), (5,8), (5,9), (6,2), (6,10), (7,10) (9,6) }

N_0 = { }
N_f = { 4 }
E = { (1,2), (1,3), (2,4), (3,4) }

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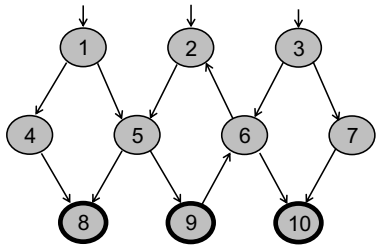
Paths in Graphs

- Path: A sequence of nodes – [n_1, n_2, …, n_M]
  - Each pair of nodes is an edge
- Length: The number of edges
  - A single node is a path of length 0
- Subpath: A subsequence of nodes in *p* is a subpath of *p*

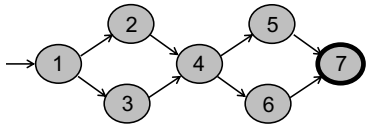


A Few Paths
[1, 4, 8]
[2, 5, 9, 6, 2]
[3, 7, 10]

7

Test Paths and SESEs

- Test Path: A path that starts at an initial node and ends at a final node
- Test paths represent execution of test cases
  - Some test paths can be executed by many tests
  - Some test paths cannot be executed by any tests
- SESE graphs : All test paths start at a single node and end at another node
  - Single-entry, single-exit (SESE)
  - N_0 and N_f have exactly one node



Double-diamond graph
Four test paths
[1, 2, 4, 5, 7]
[1, 2, 4, 6, 7]
[1, 3, 4, 5, 7]
[1, 3, 4, 6, 7]

8

Visiting and Touring

- Visit : A test path *p* *visits* node *n* if *n* is in *p*
  A test path *p* *visits* edge *e* if *e* is in *p*
- Tour : A test path *p* *tours* subpath *q* if *q* is a subpath of *p*

Path [1, 2, 4, 5, 7]
Visits nodes 1, 2, 4, 5, 7
Visits edges (1, 2), (2, 4), (4, 5), (5, 7)
Tours subpaths [1, 2, 4], [2, 4, 5], [4, 5, 7], [1, 2, 4, 5], [2, 4, 5, 7], [1, 2, 4, 5, 7]
(*Also, each edge is technically a subpath*)

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Tests and Paths

- Test criteria may require that a test starts on one node and ends at another
- Sometimes a path on a graph cannot actually be executed on a program:
  - e.g. execute a loop zero times, but the program always executes at least once (based on semantics of program)

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Tests and Test Paths

- path (*t*): The test path executed by test *t*
- path (*T*): The set of test paths executed by the set of tests *T*
- Each test executes one and only one test path
  - Complete execution from a start node to a final node
- A location in a graph (node or edge) can be reached from another location if there is a sequence of edges from the first location to the second
  - *Syntactic reach*: A subpath exists in the graph
  - *Semantic reach*: A test exists that can execute that subpath
  - This distinction will become important in section 7.3

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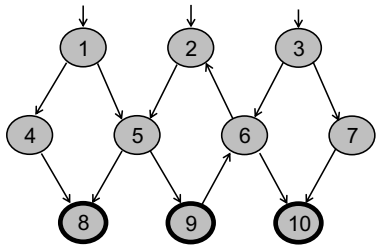
Reachability of Nodes



Can reach all nodes except 3 and 7 from Node 1

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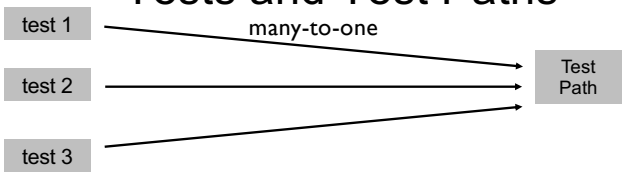
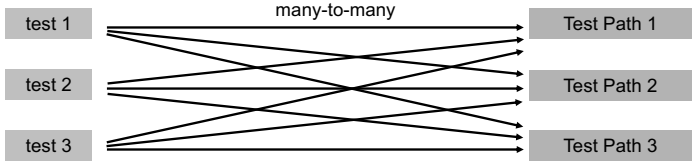
Tests and Test Paths



Deterministic software–test always executes the same test path



Non-deterministic software–the same test can execute different test paths

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Testing and Covering Graphs

- We use graphs in testing as follows :
  - Develop a model of the software as a graph
  - Require tests to visit or tour specific sets of nodes, edges or subpaths

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Testing and Covering Graphs

- Test Requirements (TR): Describe properties of test paths
- Test Criterion : Rules that define test requirements
- Satisfaction: *Given a set TR of test requirements for a criterion C, a set of tests T satisfies C on a graph if and only if for every test requirement in TR, there is a test path in path(T) that meets the test requirement tr*
- Structural Coverage Criteria: Defined on a graph just in terms of nodes and edges
- Data Flow Coverage Criteria: Requires a graph to be annotated with references to variables

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Node and Edge Coverage

- The first (and simplest) two criteria require that each node and edge in a graph be executed

Node Coverage (NC): Test set *T* satisfies node coverage on graph *G* iff for every syntactically reachable node *n* in *N*, there is some path *p* in *path(T)* such that *p* visits *n*.

- More simply stated:

Node Coverage (NC) : TR contains each reachable node in G.

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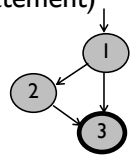
Node and Edge Coverage

- Edge coverage is slightly stronger than node

Edge Coverage (EC) : TR contains each reachable path of length up to 1, inclusive, in G.

- The phrase "*length up to 1*" allows for graphs with one node and no edges
- NC and EC are only different when there is an edge and another subpath between a pair of nodes (as in an "if-else" statement)



Node Coverage : TR = { 1, 2, 3 }
Test Path = [1, 2, 3]

Edge Coverage :TR = { (1, 2), (1, 3), (2, 3) }
Test Paths = [1, 2, 3]
[1, 3]

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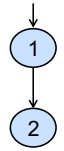
Paths of Length 1 and 0

- A graph with only one node will not have any edges

- It may seem trivial, but formally, Edge Coverage needs to require Node Coverage on this graph
- Otherwise, Edge Coverage will not subsume Node Coverage
  - So we define "length up to 1" instead of simply "length 1"
- We have the same issue with graphs that only have one edge – for Edge-Pair Coverage …



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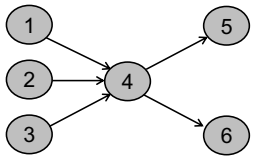
Covering Multiple Edges

- Edge-pair coverage requires pairs of edges, or subpaths of length 2

Edge-Pair Coverage (EPC) : TR contains each reachable path of length up to 2, inclusive, in G.

- The phrase "length up to 2" is used to include graphs that have less than 2 edges



Edge-Pair Coverage :

TR = { [1,4,5], [1,4,6], [2,4,5], [2,4,6], [3,4,5], [3,4,6] }

- The logical extension is to require all paths …

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Covering Multiple Edges

Complete Path Coverage (CPC) :TR contains all paths in G.

Unfortunately, this is impossible if the graph has a loop, so a weak compromise makes the tester decide which paths:

Specified Path Coverage (SPC) : TR contains a set S of test paths, where S is supplied as a parameter.

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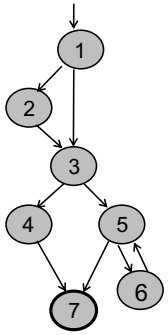
Structural Coverage Example



Node Coverage
TR = { 1, 2, 3, 4, 5, 6, 7 }
Test Paths: [1, 2, 3, 4, 7] [1, 2, 3, 5, 6, 5, 7]

Edge Coverage
TR = { (1,2), (1, 3), (2, 3), (3, 4), (3, 5), (4, 7), (5, 6), (5, 7), (6, 5) }
Test Paths: [1, 2, 3, 4, 7] [1, 3, 5, 6, 5, 7]

Edge-Pair Coverage
TR = {[1,2,3], [1,3,4], [1,3,5], [2,3,4], [2,3,5], [3,4,7], [3,5,6], [3,5,7], [5,6,5], [6,5,6], [6,5,7] }
Test Paths: [1, 2, 3, 4, 7] [1, 2, 3, 5, 7] [1, 3, 4, 7] [1, 3, 5, 6, 5, 6, 5, 7]

Complete Path Coverage
Test Paths: [1, 2, 3, 4, 7] [1, 2, 3, 5, 7] [1, 2, 3, 5, 6, 5, 6] [1, 2, 3, 5, 6, 5, 6, 5, 7] [1, 2, 3, 5, 6, 5, 6, 5, 6, 5, 7] …

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Handling Loops in Graphs

- If a graph contains a loop, it has an infinite number of paths
- Thus, CPC (complete path coverage) is not feasible
- SPC (specified coverage) is not satisfactory because the results are subjective and vary with the tester

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Handling Loops in Graphs

- Attempts to "deal with" loops:
  - 1970s : Execute cycles once ([4, 5, 4] in previous example, informal)
  - 1980s : Execute each loop, exactly once (formalized)
  - 1990s : Execute loops 0 times, once, more than once (informal description)
  - 2000s : Prime paths (touring, sidetrips, and detours)

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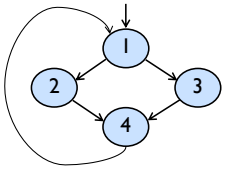
Simple Paths and Prime Paths

- Simple Path: *A path from node n_i to n_j is simple if no node appears more than once, except possibly the first and last nodes are the same*
  - No internal loops
  - A loop is a simple path
- Prime Path: *A simple path that does not appear as a proper subpath of any other simple path*



Simple Paths : [1,2,4,1], [1,3,4,1], [2,4,1,2], [2,4,1,3], [3,4,1,2], [3,4,1,3], [4,1,2,4], [4,1,3,4], [1,2,4], [1,3,4], [2,4,1], [3,4,1], [4,1,2], [4,1,3], [1,2], [1,3], [2,4], [3,4], [4,1], [1], [2], [3], [4]

Prime Paths : [2,4,1,2], [2,4,1,3], [1,3,4,1], [1,2,4,1], [3,4,1,2], [4,1,3,4], [4,1,2,4], [3,4,1,3]

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Prime Path Coverage

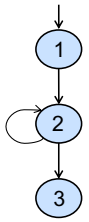
- A simple and finite criterion that requires loops to be executed as well as skipped

Prime Path Coverage (PPC) : TR contains each prime path in G.

- Will tour all paths of length 0, 1, …
- That is, it subsumes node and edge coverage
- PPC almost, but not quite, subsumes EPC …

PPC Does Not Subsume EPC

- If a node *n* has an edge to itself (*self edge*), EPC requires [*n, n, m*] and [*m, n, n*]
- [*n, n, m*] is not prime
- Neither [*n, n, m*] nor [*m, n, n*] are simple paths (not prime)



EPC Requirements :

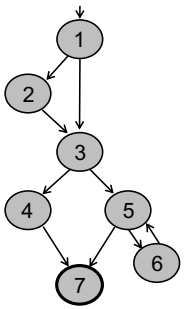
TR = { [1,2,3], [1,2,2], [2,2,3], [2,2,2] }

PPC Requirements :

TR = { [1,2,3], [2,2] }

Prime Path Example

- The previous example has 38 simple paths
- Only nine *prime paths*



Prime Paths
[1, 2, 3, 4, 7]
[1, 2, 3, 5, 7]
[1, 2, 3, 5, 6]
[1, 3, 4, 7]
[1, 3, 5, 7]
[1, 3, 5, 6]
[6, 5, 7]
[6, 5, 6]
[5, 6, 5]

Execute loop 0 times

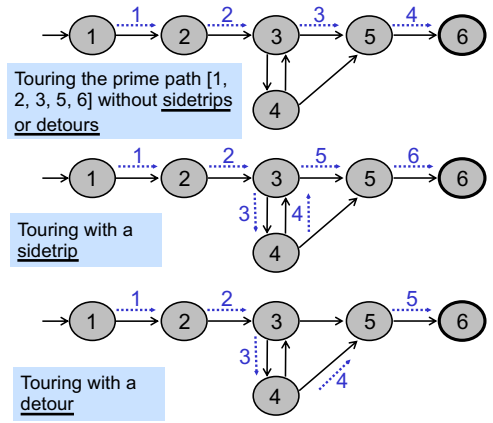
Execute loop once

Execute loop more than once

Touring, Sidetrips, and Detours

- Prime paths do not have internal loops … test paths might
- Tour: *A test path p tours subpath q if q is a subpath of p*
- Tour With Sidetrips : *A test path p tours subpath q with sidetrips iff every edge in q is also in p in the same order*
  - The tour can include a sidetrip, as long as it comes back to the same node
- Tour With Detours: *A test path p tours subpath q with detours iff every node in q is also in p in the same order*
  - The tour can include a detour from node *ni*, as long as it comes back to the prime path at a successor of *ni*

Sidetrips and Detours Example



Infeasible Test Requirements

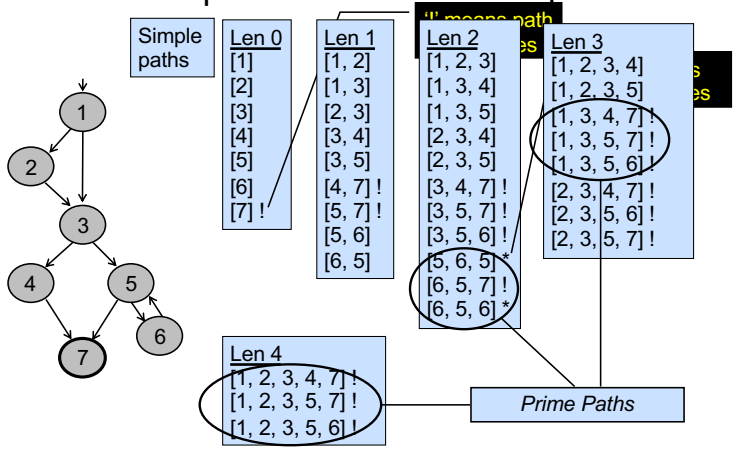
- An infeasible test requirement cannot be satisfied
  - Unreachable statement (dead code)
  - Subpath that can only be executed with a contradiction ($X > 0$ and $X < 0$)
- Most test criteria have some infeasible test requirements
- It is usually undecidable whether all test requirements are feasible

Infeasible Test Requirements

- When sidetrips are not allowed, many structural criteria have more infeasible test requirements
- However, always allowing sidetrips weakens the test criteria

Practical recommendation—Best Effort Touring
- Satisfy as many test requirements as possible without sidetrips
- Allow sidetrips to try to satisfy remaining test requirements

Simple & Prime Path Example



Round Trips

- Round-Trip Path : *A prime path that starts and ends at the same node*

Simple Round Trip Coverage (SRTC) : TR contains at least one round-trip path for each reachable node in G that begins and ends a round-trip path.

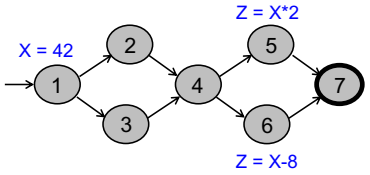
Complete Round Trip Coverage (CRTC) : TR contains all round-trip paths for each reachable node in G.

- These criteria omit nodes and edges that are not in round trips
- Thus, they do not subsume edge-pair, edge, or node coverage

Data Flow Criteria

Goal: Try to ensure that values are computed and used correctly

- Def: A location where a value for a variable is stored into memory
- Use: A location where a variable's value is accessed



Defs:	def (1) = {X}
	def (5) = {Z}
	def (6) = {Z}
Uses:	use (5) = {X}
	use (6) = {X}

The values given in defs should reach at least one, some, or all possible uses

DU Pairs and DU Paths

def (n) or def (e) : The set of variables that are defined by node n or edge e

use (n) or use (e) : The set of variables that are used by node n or edge e

DU (def-use) pair: A pair of locations (l_i, l_j) such that a variable v is defined at l_i and used at l_j

DU Pairs and DU Paths

Def-clear : A path from l_i to l_j is *def-clear* with respect to variable v if v is not given another value on any of the nodes or edges in the path

Reach : If there is a def-clear path from l_i to l_j with respect to v, the def of v at l_i reaches the use at l_j

du-path : A simple subpath that is def-clear with respect to v

from a def of v to a use of v

- du (n_i, n_j, v) – the set of du-paths from n_i to n_j
- du (n_i, v) – the set of du-paths that start at n_i

Touring DU-Paths

- A test path *p du-tours* subpath *d* with respect to *v* if *p* tours *d* and the subpath taken is def-clear with respect to *v*
- Sidetrips can be used, just as with previous touring
- Three criteria
  - Use every def
  - Get to every use
  - Follow all du-paths

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Data Flow Test Criteria

- First, we make sure every def reaches a use

All-defs coverage (ADC) : For each set of du-paths S = du (n, v), TR contains at least one path d in S.

- Then we make sure that every def reaches all possible uses

All-uses coverage (AUC) : For each set of du-paths to uses S = du (n_i, n_j, v), TR contains at least one path d in S.

- Finally, we cover all the paths between defs and uses

All-du-paths coverage (ADUPC): For each set S = du (ni, nj, v), TR contains every path d in S.

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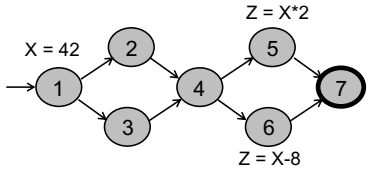
Data Flow Testing Example



All-defs for *X*
[1, 2, 4, 5]

All-uses for *X*
[1, 2, 4, 5]
[1, 2, 4, 6]

All-du-paths for *X*
[1, 2, 4, 5]
[1, 3, 4, 5]
[1, 2, 4, 6]
[1, 3, 4, 6]

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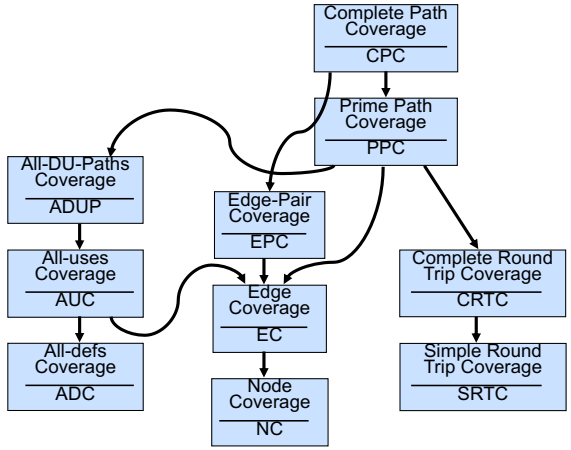
Graph Coverage Criteria Subsumption



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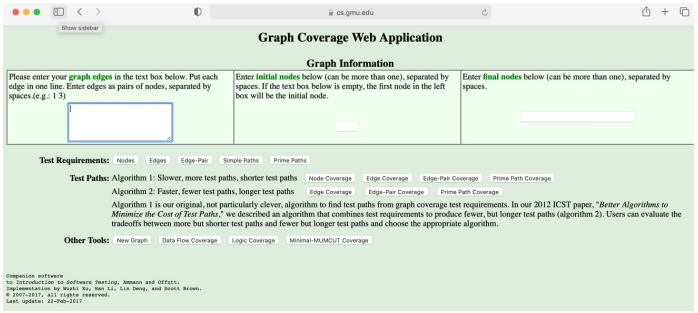
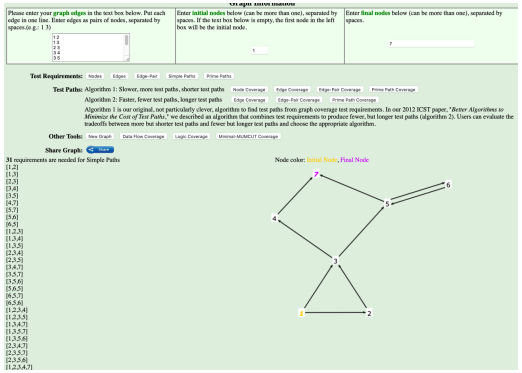
Book Website

https://cs.gmu.edu:8443/offutt/coverage/GraphCoverage



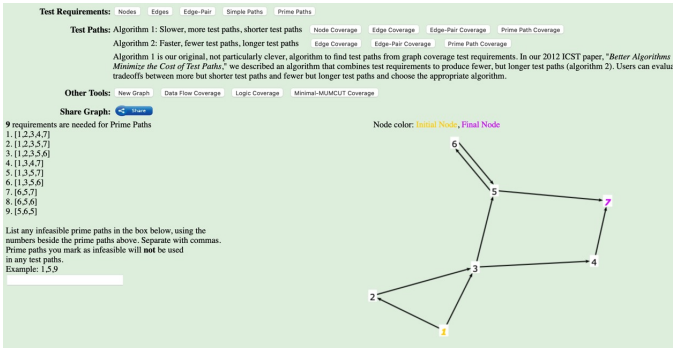
41

Book Website



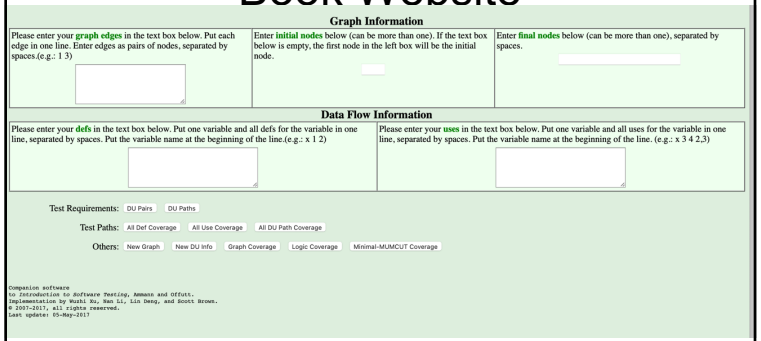
42

Book Website



43

Book Website



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For this Semester

- We have posted a temporary version (with permission from the book's authors) that wraps the original app in a python interface (with help from Claude).
- You can use when on campus (or using the Iowa State vpn.
- The app will be taken down on April 28th.

  https://mcc.apps.nimbus.las.iastate.edu/

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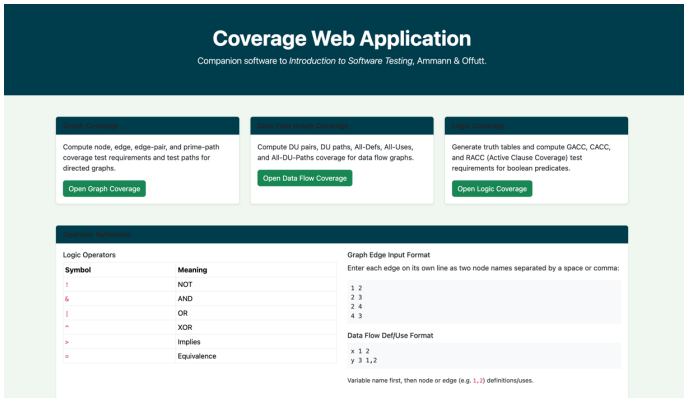
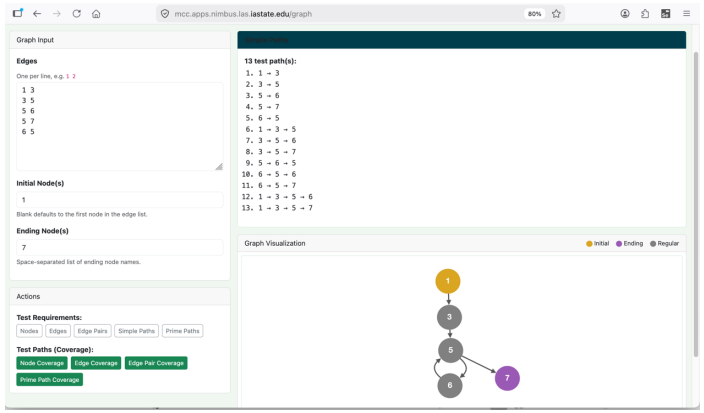


46



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